

UNEMPLOYMENT ANALYSIS WITH PYTHON

Project Report

Submitted By

Name: Aayushi kumari

Course: B.Tech Computer Science
Engineering (AI & ML)

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1. Introduction

Unemployment is one of the most important economic indicators of a country's financial health. It reflects the percentage of the labor force that is unemployed but actively seeking employment.

This project aims to analyze unemployment trends in India using Python. By examining historical unemployment data and data collected during the COVID-19 pandemic, the project identifies trends, regional differences, and the overall impact of the pandemic on employment.

2. Objectives

The objectives of this project are:

- Analyze unemployment trends in India.
- Compare unemployment rates across different states.
- Study rural and urban unemployment.
- Analyze the impact of COVID-19.

- Visualize unemployment data using charts.
- Generate meaningful insights.

3. Dataset Description

Two datasets were used in this project.

Dataset 1

File Name

Unemployment in India.csv

Contains:

- Region
- Date
- Frequency
- Estimated Unemployment Rate (%)
- Estimated Employed
- Estimated Labour Participation Rate (%)
- Area (Rural/Urban)

Dataset 2

File Name

Unemployment_Rate_upto_11_2020.csv

Contains:

- Region
- Date
- Estimated Unemployment Rate (%)
- Estimated Employed
- Estimated Labour Participation Rate (%)
- Region Group
- Latitude
- Longitude

4. Tools and Technologies

The project was developed using:

- Python
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Plotly
- Jupyter Notebook
- VS Code

5. Project Workflow

The project followed these steps:

- 1.Import Libraries
- 2.Load Dataset
- 3.Understand Dataset
- 4.Data Cleaning
- 5.Handle Missing Values
- 6.Remove Duplicate Records
- 7.Convert Date Format
- 8.Exploratory Data Analysis
- 9.Create Visualizations
10. COVID-19 Analysis

11. Generate Insights

12. Draw Conclusions

6. Data Preprocessing

The following preprocessing techniques were applied:

- Imported datasets using Pandas.
- Checked dataset dimensions.
- Examined data types.
- Identified missing values.
- Removed duplicate records.
- Converted Date column into datetime format.
- Cleaned column names.
- Created Month and Year columns for analysis.

These preprocessing steps improved data quality and prepared the dataset for analysis.

7. Exploratory Data Analysis (EDA)

Several analyses were performed, including:

- Average unemployment rate.
- Maximum and minimum unemployment rate.
- State-wise unemployment comparison.
- Average employment by state.
- Labour participation analysis.
- Rural vs Urban comparison.
- Monthly unemployment trends.
- Year-wise unemployment trends.
- COVID-19 trend analysis.

EDA helped identify significant trends and regional variations.

8. Data Visualization

The following visualizations were created:

- Histogram
- Box Plot
- Line Chart

- Bar Chart
- Pie Chart
- Scatter Plot
- Heatmap
- Pair Plot
- Monthly Trend Chart
- Year-wise Trend Chart
- State-wise Comparison
- COVID-19 Trend
- Interactive India Map using Plotly

These visualizations made it easier to understand unemployment patterns and compare regions.

9. COVID-19 Impact Analysis

The COVID-19 pandemic had a significant impact on employment across India.

The analysis showed:

- A sharp increase in unemployment during the lockdown period.
- Some states experienced much higher unemployment than others.
- Employment levels decreased during the pandemic.
- Labour participation also changed during this period.

The visualizations clearly illustrate the economic disruption caused by COVID-19.

10. Key Insights

The following insights were obtained from the analysis:

- Unemployment rates varied significantly across different states.
- COVID-19 caused a substantial rise in unemployment.

- Rural and urban unemployment patterns were different.
- Some states consistently showed higher labour participation rates.
- Employment trends changed noticeably during the pandemic.
- Regional differences indicate unequal economic impacts.

11. Conclusion

This project successfully analyzed unemployment trends in India using Python.

The project demonstrated the complete data analysis process, including data cleaning, exploratory data analysis, visualization, and interpretation of results.

The findings highlight the significant impact of COVID-19 on employment and reveal notable differences across states and regions. These insights can assist policymakers, researchers, and

analysts in understanding employment dynamics and making informed decisions.

12. Future Scope

The project can be enhanced by:

- Building machine learning models to predict unemployment rates.
- Developing an interactive dashboard using Streamlit.
- Performing time-series forecasting.
- Integrating real-time unemployment datasets.
- Deploying the project as a web application.
- Expanding the analysis with additional socioeconomic indicators.